

Designing for Creative Making in the Everyday Environment

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ABSTRACT

The development of communication and information technology has complicated the composition of space and place in the modern era, impacting human cognition and information interaction. To capture the changing nature of the information environment, I investigate information practice of arts and crafts hobbyists and their interaction with space and place. My dissertation project is grounded upon the literature on space and place in the broad information science field, regarding space and place both as meaningful social products that serve more than passive containers. Using diary studies and individual interviews, I study the adult hobbyist community without limiting the scope of discussion on where making may take place. In particular, I focus on the character of everyday space—a space that is not dedicated to support making—and how people approach and adapt to it during the actual making process. My goal is to gain a holistic understanding of spatiality by probing its influence on creative making. The findings are expected to shed new light on the design of computing technology to afford the making practice in everyday life.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

Creativity, Information interaction, Space and place, Everyday life

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1 INTRODUCTION

Making something by hand is an information rich process where space and place can be at play. Compared to the past, the advancement of technology in this era has complicated the composition of space and place and led to a more varied information environment. In such a heterogeneous information environment, people are exposed to distinct information stimuli and engage in information activities that may seem novel and unfamiliar. This condition can be particularly true in a creative making context in which creators

tend to traverse online and offline space to make use of a mix of information during the creating process. To this end, to capture the changing nature of the contemporary information environment, I investigate information practice of arts and crafts hobbyists and their interaction with space and place. Discussing their information acquisition and producing behavior, I examine how spatiality may encourage or constrain the creating practice of arts and crafts hobbyists when they undertake making projects at leisure.

The concept of space and place is often contested despite its considerable discussion in academia [1]. Tracing back to the Hellenistic period, the conversation on space and place has undergone multiple shifts and spanned a range of disciplines, such as geography, urban design, and cultural studies [2]. In the general information science field, Harrison and Dourish [3] pioneered in relating the idea of place to the development of collaborative systems. In their seminal article, Harrison and Dourish argued that rather than space, it was place, which acted as the cultural understanding of behavior and action, that could inform computer-supported cooperative work (CSCW) design. While we may design space, we could not design place since placeness arose only when we interacted with space. “Space is the opportunity; place is the understood reality” remained the motto. Ten years later, Dourish [4] published another influential paper considering the impact of mobile computing and the equal value of space and place. Drawing upon de Certeau’s perspective, Dourish posited that space and place were both meaningful and should not be deemed to be absolute dualism. As to how de Certeau delineated daily walking as “a process of appropriation of the topographic system on the part of the pedestrian” [5], Dourish considered space a social product whose production could be influenced by technology. Going one step further, Dourish and Bell [6] noted that space was embedded with different forms of infrastructures. It was these broadly defined infrastructures, such as “infrastructures of naming, infrastructures of mobility, infrastructures of separation, infrastructures of interaction,” that led our encounter with space to take on meaning. In their opinion, space is experienced as it is inhabited and is constitutive with practice. Sensitized by this view, my dissertation project explores how people approach the environment during the making process. In particular, I look into the property of everyday space—space that is not dedicated to support making—like the home. Instead of limiting the conceptual scope of spatiality, I regard space and place as multilayered and cover physical and virtual dimensions [6, 7]. Moreover, motivated by our lack of understanding of creative information practice, except for youth that is widely discussed in the literature, I focus on adult hobbyists who indulge in arts and crafts. In order to see how we may support dynamic making and afford people seamless user experiences, I aim to examine the information environment with which arts and crafts hobbyists interact during the making process. I devote attention to online and offline space and place, probing information practice of arts and crafts hobbyists to analyze information phenomena

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occurring in the environment. Throughout this research, I address the following research questions.

- How do arts and crafts hobbyists interact with space in a making context?
 - 1.1 In what type of space does the hobby take place?
 - 1.2 To what extent does space facilitate or interfere with the creating process?
 - 1.3 How do arts and crafts hobbyists construct space to make?
- What key information activities are when arts and crafts hobbyists engage in making projects?
 - 2.1 How do arts and crafts hobbyists make?
 - 2.2 What factors operate to influence the creating process?
 - 2.3 How do arts and crafts hobbyists interact with other people?

2 RESEARCH PLAN

This dissertation project is a qualitative study due to the explorative and holistic nature of my research questions. It consists of two phases: a diary study for Phase I and a semi-structured individual interview for Phase II. A general overview of the research procedure is presented in Figure 1. To begin with, I employ purposeful sampling and snowball sampling to recruit people who are over 18 years old, self-identify as an arts and crafts hobbyist, and plan to begin a project soon. Using personal contact, social media, and a campus newsletter system, I anticipate involving 20 participants as the sample size of this research. To increase data diversity, both in terms of participants' interests in arts and crafts and their creative work, I conduct an iterative sampling process with multiple rounds. In the first round of recruiting, I seek out potential participants on Behance, the largest creative network platform owned by Adobe. Behance is famous among artist and crafter communities, where both professionals and hobbyists showcase their innovative work and are inspired by others. The website has a useful search function by which I can target a specific creative field and sort the results by the most recent work to ensure that creators are still active. In the second round of recruiting, I circulate recruitment information on Reddit for its sound user base. Mainly, I select thirteen Subreddits as possible venues: DIY, Crafts, Something I Made, Pottery, Ceramics, Leathercraft, Woodworking, Metalworking, Lampwork, Stained Glass, Papercraft, Paid Study (not the same as 'Paid Research Studies,' which is for sharing market research), and Sample Size. However, this list is subject to change since each Subreddit has its moderators who may not allow a research recruitment post. In the third, also final round of recruiting, I plan to utilize a campus newsletter system to recruit people of interest. At this point, the post submitted to the newsletter system will be specially tailored to emphasize categories that have not yet or merely been slightly covered in previous rounds. People interested in participating are encouraged to contact me via email or personal message for an initial briefing session. I will share an information sheet and a diary template before each initial briefing session, during which participants can ask any questions they may have before entering Phase I. Oral Informed consent is also obtained at this stage and reconfirmed at the beginning of Phase II. Participants will be compensated for

their time and effort with \$20 Amazon electronic gift cards. To encourage people to participate in the entire study, I give participants \$15 after the first session and the rest (\$5) at the end of the second session.

For Phase I, each participant is requested to join a diary study to document the making process. I ask participants to write a diary entry every time they make progress on their projects for two weeks. There is no rule on how much time they should spend on their work or how many entries they should make throughout their participation. Participants can either extend the period or move forward to the next phase earlier. They are also free to choose on which platform they prefer to keep their diaries. Although I provide a semi-structured diary template, I allow participants to customize the layout using any artistic techniques, as long as they address the six open-ended questions below. The prompts are inspired by the protocols in previous work [8–10]. Simply put, I request participants to share the room or space where they work and something they find challenging, surprising, and exciting. There are also spaces for people to show their collected resources and jot down rambling thoughts, which are not required but highly encouraged.

1. How are you getting on with your project?
2. Where are you when working on your project? Please take photos or draw pictures of the room(s) or space(s).
3. How does the room or space look like when the arts and crafts are tidied away? If needed, please use photos or pictures to describe it.
4. Do you encounter any questions or difficulties during your making today? What are them, and how do you solve them?
5. What is something surprising during your making today?
6. How do you feel about your project so far? What is your next step?

When drafting the template, I draw upon the concept of portfolios in art and design to reflect the increasing attention on maker documentation. In particular, I combine the character of processfolios and open portfolios. According to McKay and colleagues [11], a processfolio is a traditional type of portfolio that emphasizes both a complete product and its creating process. An open portfolio is a new conceptualized category that stresses the decentralized characteristic of a portfolio by allowing the full control of makers over their documented and curated content. By bringing together the advantages of processfolios and open portfolios, I can grasp the iterative and dynamic nature of making without complicating the documenting process. For the ethical issues of diary studies, I explicitly mention in the information sheet to protect others' rights by avoiding photographing people not involved in this research [12]. If it is difficult to take photos in some situations, drawing the scene can be an alternative method for documentation. To promote autonomy in documenting and creating, I will refrain from intervening in the diary sessions and participants' making processes unless they need assistance.

Regarding Phase II, I invite each participant to join an individual interview after they wrap up the diary sessions. Integrating the interview phase into the data collection process is crucial to capturing diary content concerning space, place, and information practice, especially with a mix of textual and visual data amassed in Phase I. According to Wildemuth [13], visual data may not be adequately

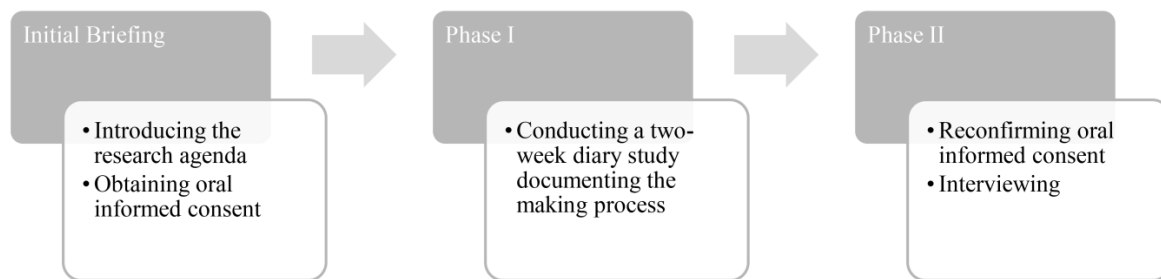


Figure 1: Research procedure.

represented if they stand alone and thereby “must be accompanied by the participant’s discussion of them and their meanings.” The interviews, running for approximately 45 minutes, are more similar to informal chat, where I request participants to debrief their work. The semi-structured interview protocol contains a few opening and closing questions, with the main emphasis on the inquiries about the creating process. Here are example questions for Phase II: “how long have you been interested in arts and crafts;” “why did you choose to document your making process in this way;” “what materials or tools did you use for your project;” “what challenges were when you managed the space for your project?” I take notes throughout these follow-up conversations to more precisely captured observed actions and emotions of participants. With their agreement, I audio record the interview sessions and later transcribe them for data analysis.

3 WORK IN PROGRESS

I am currently at the stage of data collection for my dissertation project. The research protocol was approved by the Institutional Review Board (IRB) in September 2020. I then applied for research funding from the School at the University to support my pilot study. A small pilot testing was conducted in late September and October 2020 to determine the feasibility of this project. After that, I reviewed the pilot study data and defended my dissertation proposal. I revised the recruitment strategy based on the feedback from my dissertation committee members, moving forward to the data collection of the full study in February 2021. I am now at the third round of recruiting and plan to wrap up my full study data collection by the end of August 2021. I will then begin analyzing data and produce project deliverables, expecting to defend my dissertation sometime between April and May 2022.

4 EXPECTED CONTRIBUTIONS

The expected contributions of my dissertation project are threefold. First, this research makes a theoretical contribution to space and place literature in the wide information science field by relating making with spatiality. Making is a dynamic process during which people engage in different information activities. Focusing on how creators approach online and offline space and place—i.e., the holistic information environment—without favoring space dedicated to making, I attempt to demonstrate the value of everyday space, which has often been ignored compared to the work or school setting. Previous work proved that space and place are not static

and prone to change, playing more than passive containers [1, 6]. Grounded on this perspective, I extend the discussion by probing how arts and crafts hobbyists, as a relatively under-investigated community, structure or restructure their environments as continuing making. In addition, I keep up the conversation seeing how spatiality may intervene with the making process, which may either facilitate or hamper the creative practice.

Second, my dissertation research contributes to the methodological approach to studying space and place. Although the use of diary studies is not rare in information studies, including CSCW and human-computer interaction, I employ the method openly and creatively, more akin to the artistic nature of cultural probes [14]. To offer participants extra flexibility in their creativity, I encourage people to customize the provided template by any arts-based techniques or audio journaling. Furthermore, I develop the template building on the concept of portfolios in art and design, intending to capture the individual making practice over time. Such an idea is relatively novel in the current diary study literature, which more focused on participants’ textual input.

Third, the findings of this research can make a positive contribution to the design of computing technology. The scholarly discourse on space and place has illuminated the development of pervasive computing, revealing how the computation may fit everyday life as it is removed from the desktop. However, our understanding of our daily practice may not always sufficient, especially during the pandemic period that started in early 2020. Since the outbreak, countless cities and towns worldwide were in lockdown, causing multiple physical spaces inaccessible to people. In this kind of unforeseeable situation, people had almost no choice but to switch to the online environment to keep life going, including their time at work and leisure. The change in space from physical to virtual, though it is convenient, can still disturb or complicate the routine to which individuals are familiar, even incurring their displeasure that harms the leisure pursuit. As a result, in my dissertation project, I highlight the importance of design in response to unusual circumstances without limiting the scope to a single online or offline context. Unlike other lab research, the present work is a naturalistic inquiry conducted with minimum investigator manipulation [15]. To this end, drawing upon the empirical findings, I expect to inform information professionals to think about how to harmoniously blend the online and offline information environment. Such an attempt is more than dealing with materiality of space and place but considering the cultural lens of the mundane environment. It is worth pondering how this knowledge, gained in a making context,

can be extended to other life situations, showing the potential of computation to support the spatial organization where everyday practice takes place.

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